

Confidence Intervals I

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Stat 141 | Fall 2026

Friday, Week 7

Today's Goals

- Introduce confidence intervals as a method for estimating a parameter
- Use bootstrapping as a means of creating confidence intervals

Confidence Intervals

Point Estimates

- To estimate a population parameter, we can use a sample statistic.
  - **Ex:** You're hosting a pizza party for 200 people, and need to know what proportion p of vegetarian pizza to order.
  - **Idea:** Ask a (random) sample of attendees about their pizza preference and use the proportion $\hat{p}$ in the sample as an estimate for the total proportion p.
- Your sample statistic, $\hat{p}$, is a good guess for the population parameter.
  - **Terminology:** We sometimes call our sample statistic a point estimate.

Point Estimates vs. Interval Estimates

- A sample statistic is a good guess for the population parameter, but not the whole story
  - **Q:** After polling pizza party attendees, you find $\hat{p} = 0.33$. What factor(s) determine your (un)certainty in this estimate?
- It may be preferable to estimate the proportion using **a range of values**, with smaller intervals corresponding to precision.
  - With just $n = 9$ people, you might give a range of 0.03 **to** 0.63 for p.
  - But with $n = 48$, you might instead give the range 0.20 **to** 0.46.
- We call these ranges **interval estimates**.

Confidence Interval Estimates

A **confidence interval estimate** for a parameter *usually* takes the form

$$\text{Statistic} \pm \text{Margin of Error (ME)}$$

The confidence interval gives a range of plausible values for the parameter.

- e.g., when sampling pizza preferences with $n = 48$, we estimate p using the interval

$$0.20 \text{ to } 0.46 \qquad \text{or} \qquad \underbrace{0.33}_{\text{Statistic } (\hat{p})} \pm \underbrace{0.13}_{\text{ME}}$$

The **Margin of Error** determines the width of the interval ($2 * \text{Margin of Error}$)

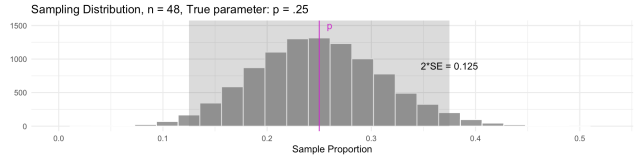
- e.g., in our pizza interval, the width is:

$$0.46 - 0.20 = 0.26 = 2 * \underbrace{0.13}_{\text{ME}}$$

Q: How should we choose our Margin of Error? (think *conceptually*)

Confidence Intervals using the Sampling Distribution

- **Goal**: Figure out a reasonable **Margin of Error**
- In our example, suppose $p = 0.25$ (i.e., 25% of all party attendees prefer vegetarian)
- For approximately bell-shaped sampling distributions, **95% of all sample statistics are within 2 SE of the parameter.**



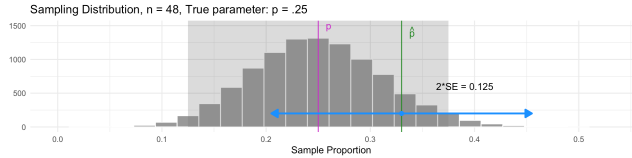
- This also means that **for 95% of all samples, the parameter will be within a distance of 2 SE of the sample statistic** (every sample in the gray region)

Confidence Intervals

Idea: Build an interval centered at the sample statistic, with a margin of error of $2 \cdot SE$:

$$\hat{p} \pm 2 \cdot \text{SE}$$

- Intervals constructed this way will contain the parameter p in 95% of samples!



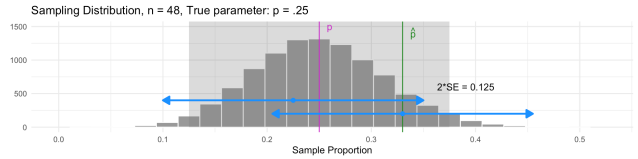
- The interval for our sample was 0.33 ± 0.125, which *does* contain the parameter p

Confidence Intervals

Idea: Build an interval centered at the sample statistic, with a margin of error of $2 \cdot SE$:

$$\hat{p} \pm 2 \cdot \text{SE}$$

- Intervals constructed this way will contain the parameter p in 95% of samples!



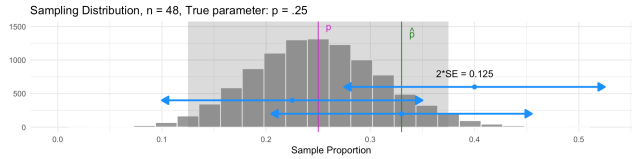
- Samples with $\hat{p}$ in the gray region have intervals that also contain the parameter p

Confidence Intervals

Idea: Build an interval centered at the sample statistic, with a margin of error of $2 \cdot SE$:

$$\hat{p} \pm 2 \cdot \text{SE}$$

- Intervals constructed this way will contain the parameter p in 95% of samples!



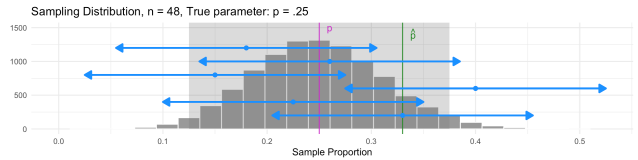
- Samples with $\hat{p}$ outside the gray region have intervals that don't contain p

Confidence Intervals

Idea: Build an interval centered at the sample statistic, with a margin of error of $2 \cdot SE$:

$$\hat{p} \pm 2 \cdot \text{SE}$$

- Intervals constructed this way will contain the parameter p in 95% of samples!

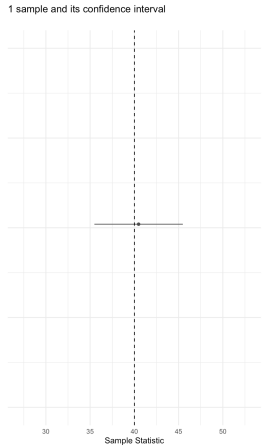


- But 95% of all samples will have intervals that do contain p

Another way to visualize this

Imagine we…

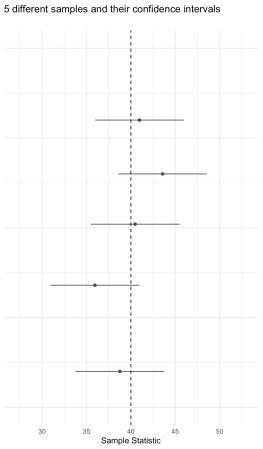
- draw a sample from our population (where the population average is 40)
- calculate the sample statistic in that sample
- construct a confidence interval for that sample using $\bar{x} \pm 2 \cdot \text{SE}$



Another way to visualize this

Imagine we…

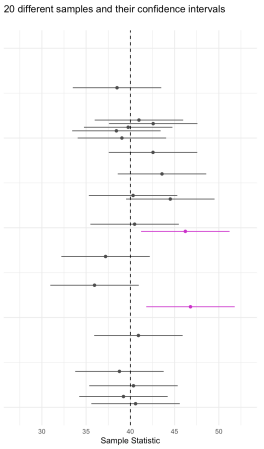
- draw 5 different samples from our population (where the population average is 40)
- calculate the sample statistic in each sample
- construct a confidence interval for each sample using $\bar{x} \pm 2 \cdot \text{SE}$



Another way to visualize this

Imagine we…

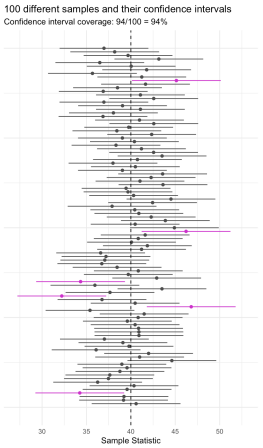
- draw 20 different samples from our population (where the population average is 40)
- calculate the sample statistic in each sample
- construct a confidence interval for each sample using $\bar{x} \pm 2 \cdot \text{SE}$



Another way to visualize this

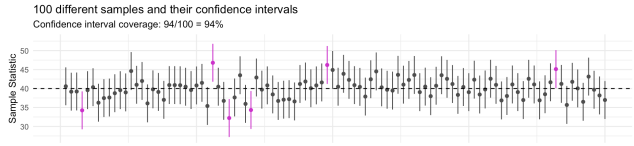
Imagine we…

- draw 100 different samples from our population (where the population average is 40)
- calculate the sample statistic in each sample
- construct a confidence interval for each sample using $\bar{x} \pm 2 \cdot \text{SE}$



Confidence levels

- **Confidence Level:** the percentage of *samples* that would yield a corresponding confidence interval containing the true value of the parameter (ex. 95%).
- For example, were we to:
  - Repeatedly draw samples from the population, and
  - Create 95% confidence intervals in each sample…
  - 95% of those confidence intervals would contain the true parameter
  - Thus, we call each interval estimate a **95% confidence interval**.
    - e.g., The pizza interval 0.33 ± 0.13 has confidence level of 95%



Recap: Confidence Intervals

- Confidence intervals consist of 1. **an interval estimate** and 2. **a confidence level**.
- In our pizza party sample, we estimated the true proportion of vegetarian pizza-eaters was between 0.20 **and** 0.46, with 95% **confidence**.
- What does "95% confidence" mean?
  - It's the success rate **of the process**
  - For 95% of all samples, the interval we construct will actually contain the parameter.
- Worth emphasizing: it's the success rate **of The Process, NOT the specific interval you calculated in your sample.**
  - The parameter is either in your interval, or it's not – there's no success rate there!

Discuss with Neighbors

Q: What's the difference between these two interpretations of a 95% confidence interval?

- For 95% of all samples, the interval we construct will actually contain the parameter.
- For a given interval, there is a 95% chance that the parameter will fall in the interval.

02:00

Interval Width and Sample Size (n)

Idea: Build a 95% confidence interval centered at the sample statistic, with a margin of error of $2 \cdot SE$:

$$\hat{p} \pm \underbrace{2 \cdot \text{SE}}_{\text{Margin of Error}}$$

- This interval's **width is determined by the Standard Error (SE).**
- **Reminder:** The SE is the standard deviation of the sampling distribution.
- **Q:** What do we know about the SE as the sample size (n) increases?
- **Q:** What does this imply about the interval's width as the sample size (n) increases?

Interval Width and Confidence Level

Idea: Build an 95% confidence interval centered at the sample statistic, with a margin of error of $2 \cdot SE$:

$$\hat{p} \pm \underbrace{2 \cdot \text{SE}}_{\text{Margin of Error}}$$

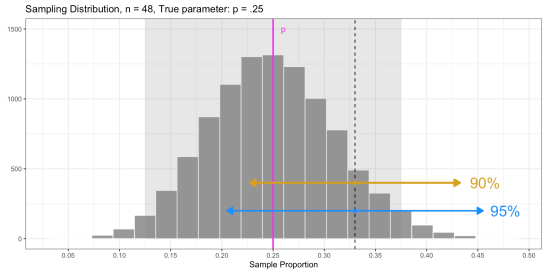
- Chose Margin of Error $= 2 * SE$ because 95% of sample statistics fall within $2 * SE$ of the parameter (in bell-shaped sampling distribution).
- **Fun Fact:** 90% of sample statistics fall within $1.64 * SE$ of the parameter.
  - **Q**: What would happen if we did the following interval?

$$\hat{p} \pm 1.64 * \text{SE}$$

  - We would get a 90% confidence interval!
  - This interval is narrower, but **has a lower "success" rate!** For 90% of all samples, the interval will contain the parameter.

Interval Width and Confidence Level

- **Takeaway:** **The confidence level also determines the width of the interval!**
  - higher confidence (e.g., 95%) means a *larger* margin of error (*wider* interval)
  - lower confidence (e.g., 90%) means a *smaller* margin of error (*narrower* interval)



Problems(?) with Confidence Intervals

Let's say we're working with 95% confidence intervals

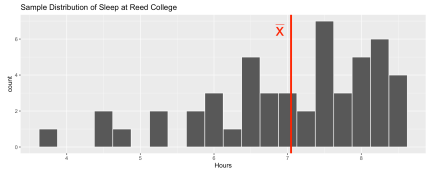
- **Problem 1:** We only have 1 sample, and we don't know if it belongs to the 95% of "good" samples, or the 5% of "bad" ones
  - **Consolation:** If I go through life constructing 95% confidence intervals, I will be right about 95% of the time
  - That's not bad!

- **Problem 2:** To make a confidence interval, we need the sampling distribution in order to compute the standard error. But in practice, we (often) don't have direct access to this.
  - **Solution:** approximate the sampling distribution via bootstrapping!

Bootstrap Confidence Intervals: Example in R

Sleep for Reed Students

Suppose we wish to estimate **how many hours Reed students sleep, on average** with a sample of 50 students, who we surveys on hours of sleep.




```
1 head(sleep, 4)
```

```
  Student    Hours
1       1 7.476953
2       2 7.717175
3       3 8.281295
4       4 7.117568
```

```
1 sleep %>% summarize(MeanSleep = round(mean(Hours), 3))
```

```
  MeanSleep
1     7.046
```

- Is the true average hours of sleep at Reed is 7.046?
  - Surely not! This is just one sample of size 50
- Let's create a *confidence interval* for the true average hours
  - We can **use the bootstrap distribution to estimate the SE** needed for the interval

Bootstrap Average Sleep

Create the bootstrap samples:

- Q: What is each argument of `rep_sample_n()` doing?
- Each bootstrap sample consists of 50 observations sampled *with replacement* from the original sample (`size = 50`)
- We have a total of 10,000 bootstrap samples (`reps = 10000`)

```
bootstrap_samples <- sleep %>%
  rep_sample_n(size = 50, replace = TRUE,
               reps = 10000)
bootstrap_samples
```

```
# A tibble: 500,000 × 3
# Groups:   replicate [10,000]
   replicate Student Hours
       <int>   <int> <dbl>
 1         1      28  4.40
 2         1      12  6.45
 3         1       7  7.71
 4         1       4  7.12
 5         1       9  7.12
 6         1       1  7.48
 7         1      27  7.26
 8         1      39  6.38
 9         1      37  3.74
10         1      44  7.98
```


Bootstrap Average Sleep

Compute bootstrap statistics: (Mean of each *bootstrap* sample)

```
bootstrap_stats <- bootstrap_samples %>%
  group_by(replicate) %>%
  summarize(x_bar = mean(Hours))
```

```
bootstrap_stats
```

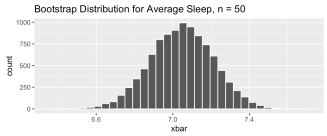
```
# A tibble: 10,000 × 2
   replicate x_bar
       <int> <dbl>
 1         1  7.05
 2         2  6.97
 3         3  6.94
 4         4  7.31
 5         5  7.36
 6         6  7.11
 7         7  6.93
 8         8  7.10
 9         9  7.06
10        10  6.76
# ℹ 9,990 more rows
```

- We now have 10,000 sample means based on the bootstrap samples, and can assess their variability

Bootstrap Average Sleep

Graph the bootstrap distribution:

```
ggplot(bootstrap_stats, aes(x = x_bar)) +
  geom_histogram(bins = 34, color = "white") +
  labs(title = "Bootstrap Distribution for Average Slee
       x = "xbar")
```



- Use the bootstrap distribution to **estimate** the standard error:

```
bootstrap_stats %>% summarize(SE = sd(x_bar))
```

```
# A tibble: 1 × 1
     SE
  <dbl>
1 0.165
```

```
# or equivalently
sd(bootstrap_stats$x_bar)
```

```
[1] 0.164749
```


Confidence Interval for Average Sleep

- Our sample average sleep was $\bar{x} = 7.046$.
- Based on the bootstrap distribution, this statistic has Standard Error = 0.165.
- Our 95% confidence interval for the true average hours of sleep for Reed students is:

$$7.046 \pm 2 \cdot 0.165$$

```
se_hat <- sd(bootstrap_stats$x_bar)
c(7.046 - 2 * se_hat, 7.046 + 2 * se_hat)
```

```
[1] 6.716502 7.375498
```

- Our best guess for average nightly sleep is that it's between 6.716 and 7.376. This method has a success rate of 95%.

Bootstrap Confidence Intervals: Complete Code Example

We are given the data frame `sleep`. Our statistic of interest is the average hours of sleep per night in the sample contained in this data frame (recorded individually as `Hours`). We can use the **bootstrap method** to calculate confidence intervals for the parameter, the average hours of sleep per night in the population:

```r
set.seed(121)
sleep <- read.csv("data-path.csv")  # Read in the data

bootstrap_samples <- sleep %>%   # Creating our 10,000 bootstrap samples
  rep_sample_n(size = 50, replace = TRUE, reps = 10000)

bootstrap_stats <- bootstrap_samples %>%  # Calculating bootstrap statistics
  group_by(replicate) %>%                 # from each sample.
  summarize(x_bar = mean(Hours))

ggplot(bootstrap_stats, aes(x = x_bar))+  # Graph the bootstrap distribution
  geom_histogram(bins = 34, color = "white")+
  labs(title = "Bootstrap Distribution for Average Sleep, n = 50", x = "xbar")

se_hat <- sd(bootstrap_stats$x_bar)  # Estimate the standard error
xbar <- mean(sleep$Hours)  # This is our **statistic**, 7.046
c(xbar - 2 * se_hat, xbar + 2 * se_hat)  # Calculate confidence interval
```


After fall break:

- More on changing confidence levels
- What to do with distributions that aren't bell-shaped (percentile method)
- Confidence interval misconceptions

Have a great fall break!